

Verification & Validation (V&V) & Benchmark Report

— BioSpecInfo

Field	Value
Software	BioSpecInfo
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Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	<code>node --check</code> on the critical JavaScript blocks on every change.	✓ In use
End-to-end (E2E) testing	Automation with Playwright on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: 0 non-network errors).	✓ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✓ In use
Manual cross-device testing	Testing on real devices (see §4).	✓ In use
Automated unit-test suite (Vitest/Jest)	—	 Roadmap
Coverage ≥ 90%	—	 Roadmap

Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ¹H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	SDBS (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	0.31 ppm
Signals within ± 0.5 ppm	89%

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	NIST Chemistry WebBook
Mean Absolute Error (MAE)	11 cm⁻¹
Bands within ± 30 cm ⁻¹	90%

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol* $\rightarrow$ $C_{27}H_{46}O$, 4 rings, 8 stereocentres.

Benchmark data source: measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30–60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30–60 fps

Size of the main `index.html` file: **≈ 4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

5. Declared model limitations

In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do **not** fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.

1. **Qualitative/educational spectra.** The simulated spectra serve to recognise functional groups and understand structure–spectrum relationships, not to replace certified experimental data.

1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.

1. **Two-way accuracy.** For rigorous use, direct links to the official databases (NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.

6. Release acceptance criteria

-  `node --check` passes without errors on the modified modules.
-  Playwright E2E: 0 non-network JS errors.
-  No visible regression in the touched sections.
-  Service Worker cache version bumped.